Machine Learning
Quiz Refresher: Chapter 5, Questions

Learning by combining models: averaging, trees, bagging and forests, feature importance, gradient boosting, XGBoost, stacking and mixtures of experts; answers in the separate answer sheet

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How to use this document

Three rounds, answers in the separate answer sheet. Round A is 20 one-line recall questions. Round B is eight multiple-choice questions, letter plus a one-line justification. Round C is five exam-style problems: the variance of an average, a Gini split, bootstrap and out-of-bag arithmetic, two rounds of gradient boosting by hand and an XGBoost leaf and split. Every number is new. Budget 20 minutes for A and B, 60 minutes for C.

Round A: rapid recall

A1. Name the three families of ensemble methods and the one-line idea of each.

A2. Hard voting versus soft voting: definitions. Which one needs calibrated models, and why?

A3. Write the variance of the average of M errors with variance σ^2 and pairwise correlation ρ. What are the two limiting cases?

A4. If every model sees the same data, where can diversity come from? Give three sources.

A5. What is a decision tree as a function? What constant does a leaf output under squared, absolute and zero–one loss?

A6. Write the Gini impurity. Its value for a pure node, and for two equally mixed classes?

A7. Write the greedy split criterion. Why "greedy"?

A8. Why are trees good base learners for ensembles? Give two reasons.

A9. What is a bootstrap sample? What fraction of distinct observations does it contain, and why that number?

A10. Define the out-of-bag prediction and the OOB error. Is the OOB error a substitute for a test set?

A11. What does a random forest add to bagging? What does $m_{\text{try}} = d$ give you?

A12. "Strength versus correlation": what does lowering m_{try} do to each?

A13. Define the mean decrease in impurity of a feature. Name its two biases.

A14. Define permutation importance. What does "masking" mean, and what can importance never establish?

A15. Write the boosting update and the initialisation for squared loss. Why is a boosted model a sum and not an average?

A16. What is a pseudo-residual? What does it reduce to under squared loss?

A17. Name the three coupled hyperparameters of boosting and the rule of thumb linking two of them. How is M chosen?

A18. For binary classification, what do we boost, and what are g and h for the cross-entropy loss?

A19. Write the Newton leaf value, the XGBoost optimal leaf weight and the gain of a split. What do λ and γ penalise?

A20. Stacking versus mixture of experts: how is the combination learned in each, and what data must the stacking meta-model be trained on?

Round B: multiple choice, letter plus justification

Each question is 2 points: 1 for the letter, 1 for a one-line justification.

B1. **Averaging.** $M = 100$ models, each with error variance $\sigma^2 = 1$, pairwise correlation $\rho = 0.5$. Variance of the averaged error:
(a) 0.01 (b) 0.505 (c) 0.5 (d) 1

B2. **Gini.** A node holds 30 examples of class 1 and 10 of class 2. Its Gini impurity:
(a) 0.375 (b) 0.5 (c) 0.25 (d) 0.625

B3. **Bootstrap.** For large N, the fraction of distinct training observations in one bootstrap sample is about:
(a) 50% (b) 36.8% (c) 63.2% (d) 100%

B4. **All the features.** Setting $m_{\text{try}} = d$ in a random forest:
(a) decorrelates the trees as much as possible (b) makes each tree see d random rows (c) is forbidden (d) turns the forest into plain bagged trees

B5. **One boosting step.** Squared loss, $F_0 = \bar{y} = 5$. An example with $y = 8$ falls in a leaf whose mean residual is 2; the learning rate is $\eta = 0.1$. Its F_1 is:
(a) 7 (b) 5.3 (c) 5.2 (d) 8

B6. **Gradient and Hessian.** Binary cross-entropy, boosted logit; an example has $\hat{p} = 0.9$ and $y = 0$. Then:
(a) $g = 0.9$, $h = 0.09$ (b) $g = -0.9$, $h = 0.09$ (c) $g = 0.1$, $h = 0.9$ (d) $g = 0.9$, $h = 0.9$

B7. **XGBoost leaf.** A leaf has $G = -6$, $H = 3$, and $\lambda = 1$. Its optimal output w^*:
(a) -2 (b) 2 (c) 1.5 (d) -1.5

B8. **Stacking gone wrong.** The meta-model is trained on the base models' predictions for their own training examples. Consequence:
(a) none, it is the standard recipe (b) the meta-model overweights the base model that overfits most, because its in-sample predictions look best (c) the meta-model underfits (d) the base models must be retrained

Round C: exam-style problems

C1. How much does averaging buy? [8 pts]

Ten regressors each make an error with variance $\sigma^2 = 4$ at some input.

(a) Variance of the averaged error when the errors are independent; when $\rho = 0.3$; when $\rho = 0.9$.

(b) With $\rho = 0.3$, what is the variance as $M \to \infty$? What is the smallest M that brings it to 1.6 or below?

(c) Three calibrated classifiers give posteriors for (class 1, class 2): $(0.6, 0.4)$, $(0.55, 0.45)$, $(0.1, 0.9)$. Hard-vote decision and soft-vote decision. Which is more defensible here, and what assumption does it rest on?

(d) Give two situations in which $\rho \approx 1$ no matter how many models you add, so that averaging cannot help.

C2. One split, two candidates [8 pts]

A node holds 40 examples: 20 positive, 20 negative. The whole training set has $N = 200$ examples. Two candidate splits:

- Split A: left $(15^+, 5^-)$, right $(5^+, 15^-)$.
- Split B: left $(10^+, 0^-)$, right $(10^+, 20^-)$.

(a) Gini impurity of the node and of each child.

(b) Weighted child impurity and impurity reduction of each split. Which does the greedy algorithm choose?

(c) Compute the credit Δ_t of the chosen split for the mean-decrease-in-impurity importance.

(d) What does each child of the chosen split predict under zero–one loss, and with what estimated posterior?

(e) The impurity reduction is largest for a feature with 500 distinct values and smallest for a binary feature, on every tree in the forest. Why is that not proof the first feature matters more? What would you compute instead?

C3. Bootstrap, out-of-bag and m_{try} [10 pts]

A random forest of $M = 500$ trees is grown on $N = 1{,}000$ examples with $d = 400$ features, of which only 5 carry signal.

(a) Expected number of distinct examples in one bootstrap sample; expected number of trees for which a given example is out-of-bag.

(b) Probability that some example is in-bag for all 500 trees (so has no OOB prediction). Is that a practical worry?

(c) The OOB error is used to pick m_{try} among $\{5, 20, 100\}$. Is the OOB error of the winner still an unbiased estimate of its test error? Is a test set still needed?

(d) With $m_{\text{try}} = 20$, estimate the probability that a given split sees at least one of the 5 informative features. Repeat for $m_{\text{try}} = 100$. Which setting do you expect to win here and why does the "$\sqrt{d}$" default fail?

(e) A tree in this forest is grown to purity on noisy labels. Why is this acceptable inside a forest but not as a standalone model?

C4. Two rounds of gradient boosting by hand [12 pts]

Regression data: $x = (1, 2, 3, 4)$, $y = (2, 4, 6, 12)$. Base learners are stumps (one split, two leaves, each leaf outputs the mean residual of its examples). Learning rate $\eta = 0.5$.

(a) F_0 and the four residuals $r_1^{(i)}$.

(b) The first stump splits at $x \leq 2$. Leaf outputs, $F_1(x^{(i)})$ for the four points and the new residuals.

(c) For the second stump compare the three possible splits ($x \leq 1$, $x \leq 2$, $x \leq 3$) by the squared error of the residual fit and pick the best. Give F_2.

(d) Training MSE of F_0, F_1, F_2.

(e) Redo (b) with $\eta = 1$. Compare F_1 and its residuals with the $\eta = 0.5$ case, and explain the rule of thumb $\eta \propto 1/M$.

(f) Why is the pseudo-residual the negative gradient of the loss, and what would it be for the absolute loss?

C5. An XGBoost leaf and whether to split it [10 pts]

Binary classification, boosted logit, cross-entropy loss. A leaf holds three examples with (label, current probability): $(1, 0.3)$, $(1, 0.6)$, $(0, 0.2)$.

(a) Compute $g^{(i)}$ and $h^{(i)}$ for each example, then G and H.

(b) Optimal leaf weight w^* for $\lambda = 0$ and for $\lambda = 1$. Loss removed by the leaf for $\lambda = 1$.

(c) A candidate split sends the first two examples left and the third right. Compute G_L, H_L, G_R, H_R and the gain with $\lambda = 1$, $\gamma = 0.1$. Split or not? Repeat with $\gamma = 0.2$.

(d) Explain why the second-order step gives a closed-form leaf value while the first-order (plain gradient) step does not.

(e) Three practitioners each change one of η, the depth and M in isolation to fix overfitting. Say in one sentence each what goes wrong, and what the right procedure is.